CUSTOMER SHOPPING BEHAVIOR ANALYSIS

Project Overview

The **Customer Behavior Analysis** project is designed to analyze a dataset containing 3,900 customer records across 18 variables, covering customer demographics, purchase details, and shopping behavior. The dataset includes key attributes such as age, gender, location, subscription status, product categories, purchase amounts, seasonal trends, and customer engagement indicators like discounts, promo code usage, and purchase frequency.

The primary objective of this project is to uncover patterns in customer purchasing behavior and identify factors that influence revenue generation and customer retention. Using SQL for data querying and Power BI for visualization, the analysis explores relationships between customer demographics and spending habits, evaluates product performance, and assesses the effectiveness of marketing strategies such as discounts and subscription models.

Additionally, the project addresses data quality by handling missing values (notably 37 missing entries in the review rating column) to ensure accurate insights. It also segments customers based on their purchase history to distinguish between new, returning, and loyal customers.

The final outcome is an interactive dashboard that provides actionable insights to support business decisions, optimize customer targeting, enhance user experience, and improve overall sales performance.

EDA Using Python

● Data Loading: Imported the dataset using pandas.

● Initial Exploration: Used df.info() to check structure and .describe() for summary statistics.

● Missing Data Handling: Checked for null values and imputed missing values in the Review Rating column using the median rating of each product category.

● Column Standardization: Renamed columns to snake case for better readability and documentation.

● Feature Engineering: Created age_group column by binning customer ages and Created purchase_frequency_days column from purchase data.

● Data Consistency Check: Verified if discount_applied and promo_code_used were redundant; dropped promo_code_used.

● Database Integration: Connected Python script to PostgreSQL and loaded the cleaned DataFrame into the database for SQL analysis.

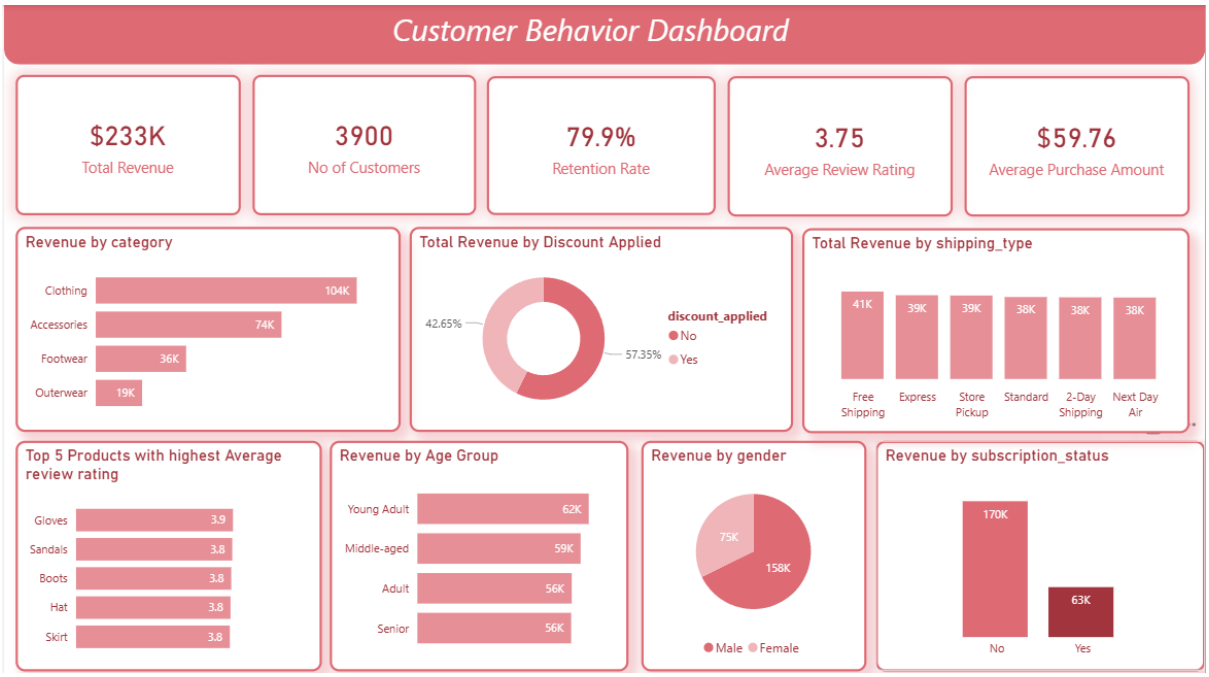
Data Analysis using SQL

1. Revenue by Gender – Compared total revenue generated by male vs. female customers.
2. High-Spending Discount Users – Identified customers who used discounts but still spent above the average purchase amount.
3. Top 5 Products by Rating – Found products with the highest average review ratings.
4. Shipping Type Comparison – Compared average purchase amounts between Standard and Express shipping.

5. Subscribers vs. Non-Subscribers – Compared average spend and total revenue across subscription status.
6. Discount-Dependent Products – Identified 5 products with the highest percentage of discounted purchases.
7. Customer Segmentation – Classified customers into New, Returning, and Loyal segments based on purchase history.
8. Top 3 Products per Category – Listed the most purchased products within each category.
9. Repeat Buyers & Subscriptions – Checked whether customers with >5 purchases are more likely to subscribe.
10. Revenue by Age Group – Calculated total revenue contribution of each age group.

Power BI Dashboard

An interactive **Customer Behavior Dashboard** was developed using Power BI to analyze key business metrics related to customer purchasing patterns, revenue distribution, and retention behavior. The dashboard provides a comprehensive overview of customer activity through well-structured KPIs and visualizations.



Key Performance Indicators (KPIs)

At the top of the dashboard, key metrics are displayed to provide a quick summary of overall performance:

- **Total Revenue:** $233K, indicating overall sales performance
- **Number of Customers:** 3,900, representing the customer base size
- **Retention Rate:** 79.9%, highlighting strong customer loyalty
- **Average Review Rating:** 3.75, reflecting customer satisfaction levels
- **Average Purchase Amount:** $59.76, showing typical customer spending

---

Revenue Analysis

- **Revenue by Category:**
  Clothing generates the highest revenue, followed by accessories, footwear, and outerwear, indicating strong demand in apparel products.
- **Revenue by Discount Applied:**
  A significant portion of revenue comes from discounted purchases, showing that promotional strategies play an important role in driving sales.
- **Revenue by Shipping Type:**
  Revenue is fairly distributed across shipping methods, with free shipping contributing slightly higher, suggesting customer preference for cost-saving delivery options.

Customer & Demographic Insights

- **Revenue by Age Group:**
  Young adults contribute the highest revenue, followed by middle-aged and adult groups, indicating that younger customers are more active buyers.
- **Revenue by Gender:**
  Male customers contribute a larger share of revenue compared to female customers, highlighting a potential target segment.
- **Revenue by Subscription Status:**
  Non-subscribed customers generate more revenue than subscribed customers, suggesting an opportunity to improve subscription engagement strategies.

Product Performance

- **Top 5 Products by Average Rating:**
  Products like gloves, sandals, boots, hats, and skirts have the highest ratings, indicating strong customer satisfaction and potential for promotion.

Overall Insights

The dashboard effectively highlights customer behavior trends, showing that:

- Apparel categories dominate sales performance
- Discounts significantly influence purchasing decisions
- Younger customers are key contributors to revenue
- Customer retention is relatively high, indicating strong engagement
- Subscription models require optimization to increase value

Business Recommendations

- Boost Subscriptions – Promote exclusive benefits for subscribers.
- Customer Loyalty Programs – Reward repeat buyers to move them into the “Loyal” segment.
- Review Discount Policy – Balance sales boosts with margin control.
- Product Positioning – Highlight top-rated and best-selling products in campaigns.
- Targeted Marketing – Focus efforts on high-revenue age groups and express-shipping users.